A Counterexample to a Hitting-Time Necessity Question for Random Walks on Growing Graphs

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Abstract

Kijima, Shimizu, and Shiraga [1] asked whether, for arbitrary random walks on growing graphs,

$$\mathbb{E}[U(n)] = O(1) \quad \Longrightarrow \quad d(i) = \Omega\big(t_{\mathrm{hit}}(i)\big).$$

This note gives a negative answer in the stated generality. Thus the result is a solution to that general question, not merely partial progress. The counterexample uses complete graphs and lazy reversible transition matrices. The duration is only linear, $d(i) = \lceil 8i \rceil$, while the worst-case hitting time is at least $2i^3$. Nevertheless, the expected number of unvisited vertices remains uniformly bounded, because the unique very hard-to-hit vertex is the initial vertex and hence is already visited at time zero. The example does not address stronger variants in which the transition matrices are required to be simple, symmetric, or to have the uniform stationary distribution.

1 Model and statement of the question

We use the notation of random walks on growing graphs. For each $i \geq 1$, let

$$G^{(i)} = (V^{(i)}, E^{(i)}), \qquad V^{(i)} = \{v_1, \ldots, v_i\},$$

be a connected graph, with $G^{(i+1)}$ obtained from $G^{(i)}$ by adding the new vertex v_{i+1} and at least one incident edge. A duration function $d : \mathbb{N} \to \mathbb{N}$ specifies that the process spends exactly $d(i)$ random-walk steps on $G^{(i)}$. Let

$$T_1 := 0, \qquad T_i := \sum_{j=1}^{i-1} d(j) \quad (i \geq 2).$$

During the interval in which the current graph is $G^{(i)}$, the walk uses a transition matrix $P^{(i)}$ on $V^{(i)}$. We write $(Z_t)_{t\geq 0}$ for the resulting time-inhomogeneous walk.

For a fixed matrix $P^{(i)}$, define the worst-case hitting time

$$t_{\mathrm{hit}}(i) := \max_{u,v \in V^{(i)}} \mathbb{E}_u^{(i)}[\tau_v], \qquad \tau_v := \min\{t \geq 0 : X_t = v\},$$

where (X_t) is the time-homogeneous Markov chain with transition matrix $P^{(i)}$. The number of vertices of $G^{(n)}$ not yet visited just before the next vertex is added is

$$U(n) := \left| V^{(n)} \setminus \bigcup_{t=0}^{T_{n+1}} \{Z_t\} \right|.$$

This is the question stated in the authors' paper and recorded on the Open Problems in Operations Research page [2]. The question is whether $\mathbb{E}[U(n)] = O(1)$ forces $d(i) = \Omega(t_{\mathrm{hit}}(i))$ for every random walk on a growing graph. We prove that it does not.

2 The construction

For every $i \geq 1$, let $G^{(i)} = K_i$, the complete graph on $\{v_1, \dots, v_i\}$. Let

$$d(i) := \lceil 8i \rceil.$$

The transition matrices for $i = 1, 2$ are chosen as follows. Let $P^{(1)}(v_1, v_1) = 1$. For $i = 2$, set

$$P^{(2)}(v_a, v_b) = \frac{1}{2} \qquad (a, b \in \{1, 2\}).$$

For $i \geq 3$, put

$$R_i := \{v_2, \dots, v_i\}, \qquad p_i := i^{-3}.$$

The vertex v_1 will be the special vertex. From v_1, define

$$P^{(i)}(v_1, v_1) = \frac{1}{2}, \qquad P^{(i)}(v_1, r) = \frac{1}{2(i-1)} \quad (r \in R_i).$$

From a regular vertex $r \in R_i$, define

$$P^{(i)}(r, r) = \frac{1}{2}, \qquad P^{(i)}(r, v_1) = \frac{p_i}{2}, \qquad P^{(i)}(r, s) = \frac{1 - p_i}{2(i-2)} \quad (s \in R_i,\ s \neq r).$$

All non-diagonal positive transitions are along edges of K_i, and the diagonal entries encode laziness.

Proposition 1. *For every i, the matrix $P^{(i)}$ is a stochastic irreducible transition matrix on K_i. For $i \geq 3$, it is lazy and reversible with stationary distribution*

$$\pi_i(v_1) = \frac{p_i}{1 + p_i}, \qquad \pi_i(r) = \frac{1}{(i-1)(1+p_i)} \quad (r \in R_i).$$

Proof. The cases $i = 1, 2$ are immediate. Suppose $i \geq 3$. The row sum at v_1 is

$$\frac{1}{2} + (i-1)\frac{1}{2(i-1)} = 1.$$

For $r \in R_i$, the row sum is

$$\frac{1}{2} + \frac{p_i}{2} + (i-2)\frac{1-p_i}{2(i-2)} = 1.$$

Thus $P^{(i)}$ is stochastic. It is irreducible because every regular vertex has positive transition probability to v_1, and v_1 has positive transition probability to every regular vertex. It is lazy because all diagonal entries are $1/2$.

The displayed formula for π_i is a probability distribution, since

$$\frac{p_i}{1+p_i} + (i-1)\frac{1}{(i-1)(1+p_i)} = 1.$$

Detailed balance holds between v_1 and any $r \in R_i$, because

$$\pi_i(v_1)P^{(i)}(v_1, r) = \frac{p_i}{1+p_i} \cdot \frac{1}{2(i-1)} = \frac{p_i}{2(i-1)(1+p_i)} = \frac{1}{(i-1)(1+p_i)} \cdot \frac{p_i}{2} = \pi_i(r)P^{(i)}(r, v_1).$$

For distinct $r, s \in R_i$, the stationary masses are equal and $P^{(i)}(r, s) = P^{(i)}(s, r)$. The diagonal cases are automatic. Hence $P^{(i)}$ is reversible. $\square$

3 Uniformly bounded expected number of missed vertices

We first record the elementary conditional-hazard estimate used below.

Lemma 1. *Let a stage have length m. Suppose a vertex w exists throughout the stage and, at each step of the stage, conditional on the history so far and on w not yet having been visited, the probability of visiting w on the next step is at least q. Then the probability that w is not visited during that stage, conditional on being unvisited at the start of the stage, is at most $(1-q)^m$.*

Proof. Let A_j be the event that w remains unvisited after the first j steps of the stage, with the conditioning on the past before the stage understood. The hypothesis gives

$$\mathbb{P}(A_{j+1} \mid A_j, \mathcal{F}_j) \le 1-q$$

whenever A_j occurs. Taking conditional expectations iteratively gives

$$\mathbb{P}(A_m) \le (1-q)^m \, \mathbb{P}(A_0) = (1-q)^m,$$

under the condition that w was unvisited at the start of the stage. □

Lemma 2. *For the construction above,*

$$\sup_{n\ge 1} \mathbb{E}[U(n)] < \infty.$$

More explicitly, for all $n \ge 3$,

$$\mathbb{E}[U(n)] \le 1 + \frac{e^{-2}}{1-e^{-2}}.$$

Proof. The vertex v_1 is visited at time 0, so it never contributes to $U(n)$. Fix $k \ge 3$. Consider a stage $i \ge k$. Conditional on the event that v_k has not yet been visited, the current state is not v_k. If the current state is v_1, then the next-step probability of moving to v_k is

$$P^{(i)}(v_1, v_k) = \frac{1}{2(i-1)} \ge \frac{1}{4i}.$$

If the current state is a regular vertex in $R_i \setminus \{v_k\}$, then the next-step probability of moving to v_k is

$$\frac{1-p_i}{2(i-2)} \ge \frac{1}{4i}.$$

Indeed, for $i \ge 3$, the inequality follows from $p_i = i^{-3} \le 1/2$, since

$$\frac{1-p_i}{2(i-2)} \ge \frac{1}{4(i-2)} \ge \frac{1}{4i}.$$

Thus, throughout stage i, as long as v_k is still unvisited, the conditional probability of hitting v_k at the next step is at least $1/(4i)$. By Lemma 1 and $d(i) = \lceil 8i \rceil$,

$$\begin{aligned}\mathbb{P}\{v_k \text{ is not hit during stage } i \mid v_k \text{ is unvisited at the start of stage } i\} &\le \left(1 - \frac{1}{4i}\right)^{d(i)} \\ &\le \exp\left(-\frac{d(i)}{4i}\right) \le e^{-2}.\end{aligned}$$

Iterating over the stages $i = k, k+1, \ldots, n$ gives

$$\mathbb{P}\{v_k \text{ is unvisited by time } T_{n+1}\} \le e^{-2(n-k+1)} \qquad (3 \le k \le n).$$

Therefore, for $n \geq 3$,

$$\begin{aligned}\mathbb{E}[U(n)] &= \sum_{k=1}^{n} \mathbb{P}\{v_k \text{ is unvisited by time } T_{n+1}\} \\ &\leq 1 + \sum_{k=3}^{n} e^{-2(n-k+1)} \\ &\leq 1 + \sum_{m=1}^{\infty} e^{-2m} = 1 + \frac{e^{-2}}{1-e^{-2}}.\end{aligned}$$

Here the term 1 is a crude bound for the possible contribution of v_2; the contribution of v_1 is zero. The finitely many cases $n < 3$ are absorbed by increasing the constant. □

4 The worst-case hitting time is much larger than the duration

Lemma 3. *For every $i \geq 3$,*

$$t_{\text{hit}}(i) \geq 2i^3.$$

Proof. Fix $i \geq 3$ and start the chain with transition matrix $P^{(i)}$ from any regular vertex $r \in R_i$. Until the chain hits v_1, it remains inside R_i. From every state in R_i, the one-step probability of jumping to v_1 is exactly

$$\frac{p_i}{2} = \frac{1}{2i^3}.$$

Consequently,

$$\mathbb{P}_r^{(i)}(\tau_{v_1} > m) = \left(1 - \frac{p_i}{2}\right)^m \qquad (m \geq 0),$$

and τ_{v_1} is a geometric random variable on $\{1, 2, \ldots\}$ with success probability $p_i/2$. Therefore

$$\mathbb{E}_r^{(i)}[\tau_{v_1}] = \frac{2}{p_i} = 2i^3.$$

Since $t_{\text{hit}}(i)$ is the maximum over all ordered pairs of starting and target vertices, $t_{\text{hit}}(i) \geq 2i^3$. □

5 Conclusion

Theorem 1. *There exists a random walk on growing graphs such that*

$$\sup_{n\geq 1} \mathbb{E}[U(n)] < \infty, \qquad \textit{but} \qquad \frac{d(i)}{t_{\text{hit}}(i)} \longrightarrow 0.$$

The implication

$$\mathbb{E}[U(n)] = O(1) \quad \Longrightarrow \quad d(i) = \Omega\big(t_{\text{hit}}(i)\big)$$

is therefore false for arbitrary random walks on growing graphs.

Proof. The construction of Proposition 1 is an admissible random walk on growing complete graphs. By Lemma 2, its expected number of unvisited vertices is uniformly bounded. By Lemma 3, for $i \geq 3$,

$$0 \leq \frac{d(i)}{t_{\text{hit}}(i)} \leq \frac{\lceil 8i \rceil}{2i^3} \leq \frac{8i+1}{2i^3} \longrightarrow 0.$$

This proves the asserted separation. □

Remark 1 (Scope of the counterexample). *The construction exploits the difference between worst-case hitting time and the probability of hitting vertices that are actually still unvisited. The worst target is v_1, because it has stationary mass of order i^{-3} and is hit from regular vertices only with probability $i^{-3}/2$ per step. But v_1 is the initial vertex, so it is already covered at time zero. Every newly added vertex lies in R_i and is hit during stage i with hazard of order $1/i$, which linear duration controls. Thus the example gives a full negative solution to the general question, while leaving open any strengthened version that forbids this mechanism, such as a version restricted to simple random walks, symmetric transition matrices, or uniform stationary distributions.*

References

[1] S. Kijima, N. Shimizu, and T. Shiraga, *How Many Vertices Does a Random Walk Miss in a Network with a Moderately Increasing Number of Vertices?* Mathematics of Operations Research, 51(1):783–805, 2026. DOI: 10.1287/moor.2023.0060. Preprint version: arXiv:2008.10837.

[2] Open Problems in Operations Research, *Characterize when constant expected unvisited vertices requires duration at least hitting time*, problem W3080982993_p0. `https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W3080982993_p0&page=6&n=58`.